Magnitude, Distance and Scalar Operations on Vectors

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Where this fits: Pipeline map step 0 (Foundations). See the Course map.

0 Foundations	1 Frame	2 Get data	3 Understand	4 Clean	5 Features	6 Reduce
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- **Builds on:** Vectors and feature vectors.
- **Leads to:** Regularisation; K-nearest neighbours; Heuristic; Growing obstacles by the robot's radius; Metric.
- **Compare with:** Cosine similarity.

1. Overview

Key point: Pythagoras' theorem gives both the length of a vector and the distance between two vectors, in any number of dimensions; adding or multiplying by a **scalar** (G-1743) shifts or scales a vector.

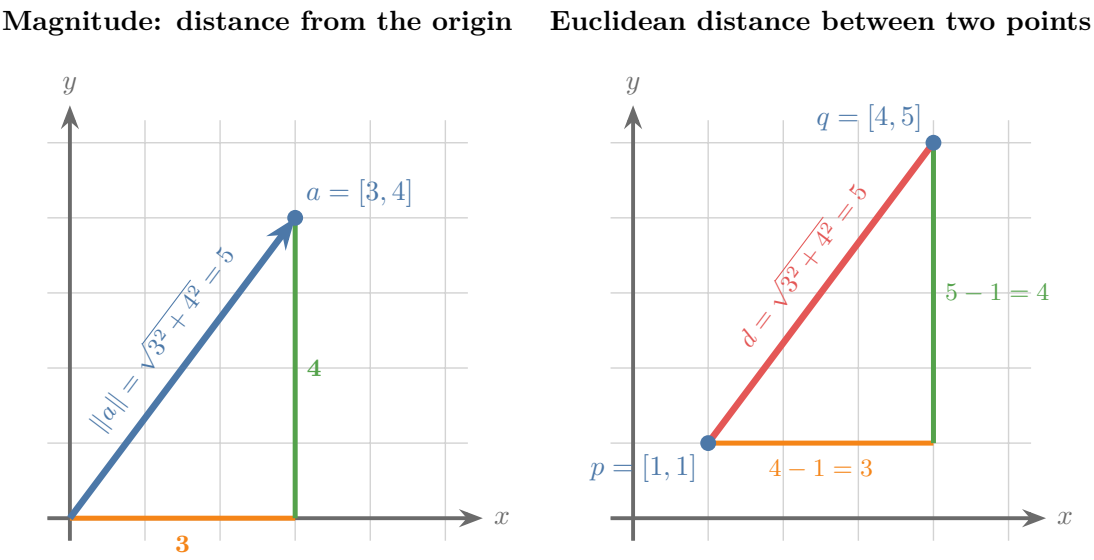


Figure 1: Left: the length of a vector. Right: the distance between two points. Both are Pythagoras' theorem

Figure 1 shows the two measurements this Note builds on. The length of $[3, 4]$ and the distance from $[1, 1]$ to $[4, 5]$ are both 5, and both come from the same right triangle. This Note:

- works out the length in 2D, 3D and n dimensions (section 2);
- does the same for the distance, and shows where ML uses it (section 3);
- turns to the simplest operation on a vector, adding a scalar (section 4), and its main use in ML, **mean centring** (G-1195) (section 5);
- ends with multiplying by a scalar (section 6).

A **vector** is a list of numbers such as $[3, 4]$; each number is a **component**, and the number of components is the vector's **dimension**. A **scalar** is a single number such as 5. All are defined in what a vector is.

2. Magnitude: the distance from the origin

> **Key point:** The magnitude of a vector is the square root of the sum of its squared components; the same formula works in every dimension.

The **magnitude** of a vector is its distance from the origin: the length of the arrow from its tail to its head. The magnitude is also called the length or the norm of the vector (the name used in the margin in terms of w), and written $\|x\|$.

2.1 In 2D and 3D

> **Key point:** In 2D the two components are the legs of a right triangle, so Pythagoras gives the length.

In Figure 1 (left), the vector $a = [3, 4]$ forms a right triangle with the x-axis: one leg is 3 long, the other 4. The arrow is the hypotenuse, so by Pythagoras' theorem its length is:

$$\sqrt{3^2 + 4^2} = \sqrt{25} = 5$$

For a general 2D vector $[a, b]$ the length is:

$$\sqrt{a^2 + b^2}$$

The same logic holds in 3D, with Pythagoras used twice (Figure 2). For $[2, 3, 6]$, the floor diagonal under the arrow joins $[0, 0, 0]$ to $[2, 3, 0]$ and has this length:

$$\sqrt{2^2 + 3^2} = \sqrt{13}$$

That diagonal and the height 6 are the legs of a second right triangle, whose hypotenuse is the arrow:

$$\sqrt{13 + 36} = \sqrt{49} = 7$$

In general a vector $[a, b, c]$ has this length:

$$\sqrt{a^2 + b^2 + c^2}$$

It is the square root of the sum of the squares of all its components.

$$\|[2,\,3,\,6]\| = \sqrt{(2^2 + 3^2 + 6^2)} = \sqrt{49} = 7$$

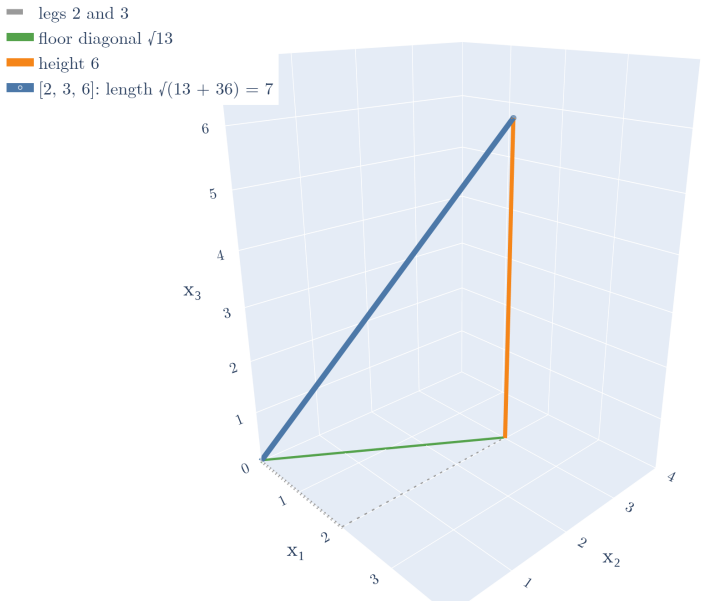


Figure 2: The length of [2, 3, 6] by Pythagoras twice: the floor diagonal (green) has length the square root of 13; with the height 6 (orange) it forms a right triangle whose hypotenuse, the vector (blue), has length 7.

2.2 In n dimensions

> **Key point:** Square every component, add, take the square root.

A ruler measures an arrow drawn in 2D or 3D. A vector with 50 components cannot be drawn, so its length has to come from a rule, and the rule must agree with Pythagoras wherever we can draw.

1. **In words:** square every component, add the squares, and take the square root.
2. **Formula:** for $x = [x_1, x_2, \ldots, x_n]$,

$$\|x\| = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}$$

$$\|x\| = \sqrt{\sum_{i=1}^{n} x_i^2}$$

   Here $\sum_{i=1}^{n}$ means: add the terms for $i = 1, 2, \ldots, n$.

3. **Example:** for the 3D vector $[2, 3, 6]$,

$$\|x\| = \sqrt{4 + 9 + 36} = \sqrt{49} = 7$$

   and for the 5D vector $[1, 2, 3, 4, 5]$,

$$\|x\| = \sqrt{1 + 4 + 9 + 16 + 25}$$

$$\|x\| = \sqrt{55} \approx 7.42$$

The sum under the square root is the vector multiplied with itself, component by component. That sum is the dot product of the vector with itself, taught in computing the dot product, so:

$$\|x\|^2 = x \cdot x$$

For $[2, 3, 6]$:

$$2 \times 2 + 3 \times 3 + 6 \times 6$$

$$4 + 9 + 36 = 49 = 7^2$$

> **Python:** NumPy computes the magnitude of a vector of any dimension with one function.
> ```
> import numpy as np
>
> v = np.array([1, 2, 3, 4, 5])
> np.linalg.norm(v)    # 7.416...
> ```
> `np.linalg` is NumPy's linear algebra module. The same call works unchanged for a 15-dimensional or a 500-dimensional vector.

> **Extra:** The magnitude is one norm among several, the **L2 norm** (G-1028); adding the absolute values instead, $|x_1| + \cdots + |x_n|$, gives the **L1 norm** (G-1025). Ridge regression penalises the squared L2 norm of the coefficients and Lasso their L1 norm (see Ridge penalising large coefficients and Lasso on one feature), so vector norms are the building block of regularisation.

3. Euclidean distance

> **Key point:** The Euclidean distance between two vectors is the magnitude of their difference: subtract component by component, then take the norm.

The **Euclidean distance** (G-715), the straight-line distance between two points, is defined in the Euclidean distance. Here we see how it links to the magnitude.

3.1 Distance as the norm of the difference

> **Key point:** Subtract the two vectors, then take the magnitude of the result.

In Figure 1 (right) the legs of the triangle, 3 and 4, are the components of the difference vector $q - p$:

$$q - p = [4, 5] - [1, 1]$$

$$q - p = [3, 4]$$

The distance is the hypotenuse, which is exactly the magnitude of that vector. Subtracting the other way round, $p - q = [-3, -4]$, only flips the signs, and squaring removes them, so the order does not matter:

$$d(p, q) = \|p - q\|$$

The squared differences in the distance formula are the squared components of $p - q$. So for $p = [6, 7, 8, 9, 10]$ and $q = [1, 2, 3, 4, 5]$ the difference is $[5, 5, 5, 5, 5]$ and the distance is the square root of five squares of 5, one step per line:

$$5^2 + 5^2 + 5^2 + 5^2 + 5^2 = 125$$

$$\sqrt{125} \approx 11.18$$

> **Python:** The distance is two steps: subtract, then `norm`.
> ```
> p = np.array([6, 7, 8, 9, 10])
> q = np.array([1, 2, 3, 4, 5])
> np.linalg.norm(p - q)    # 11.18...
> ```
> This works for vectors of any dimension, as long as both have the same number of components.

3.2 Distance in ML: the nearest neighbour

> **Key point:** Many algorithms decide by distance: a new point gets the class of the vectors nearest to it.

A lot of ML algorithms compute **Euclidean distances** (G-715) inside. The clearest example is **K-nearest neighbours** (G-998) (KNN, G-998), a classification algorithm. To classify a new iris flower from its sepal length and petal length, KNN treats the flower as a vector, computes its distance to every flower in the data, and gives it the species of the nearest ones.

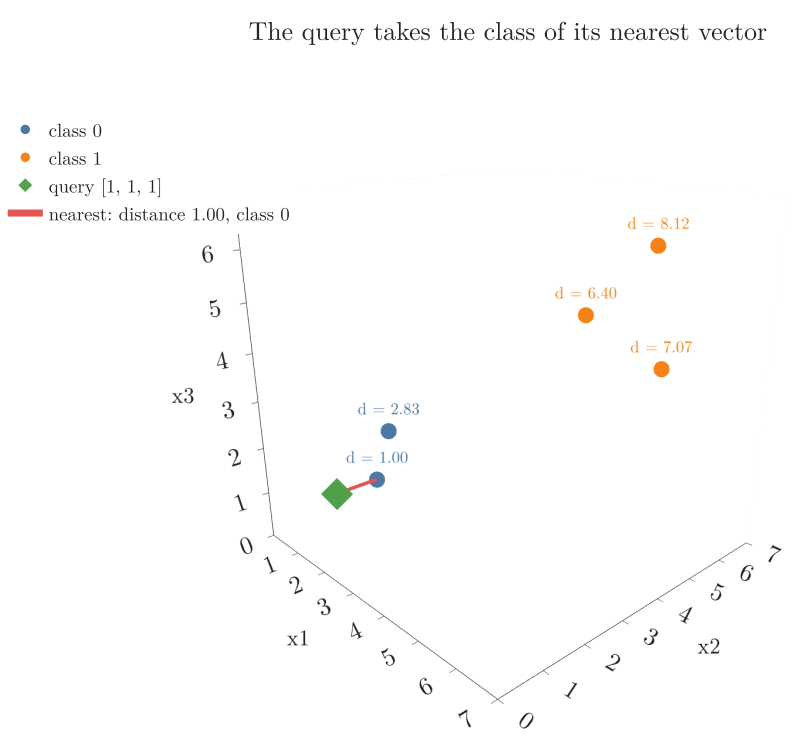


Figure 3: Five labelled vectors in 3D and a query vector; the query takes the class of the nearest one

Figure 3 runs this on five 3D vectors in two classes. The query vector $[1, 1, 1]$ is 1.00 from $[1, 2, 1]$ and at least 2.83 from all others. Its nearest neighbour is in class 0, so the query is classed 0.

The same distance appears in K-means clustering, which groups points around their nearest centre, and in recommender systems, as in recommending by distance.

4. Operations with a scalar

> **Key point:** A **scalar** (G-1743) operation applies the same number to every component; adding or subtracting a scalar shifts the vector.

A vector and a scalar can be combined in two ways: by adding (or subtracting, which works the same way) and by multiplying (or dividing). In both, the scalar is applied to every component in turn.

4.1 Adding or subtracting a scalar: shifting

> **Key point:** Adding a **scalar** (G-1743) adds it to every component, which moves the point; this is called shifting.

1. **In words:** add the scalar to every component of the vector.
2. **Formula:** for $v = [v_1, v_2, \ldots, v_n]$ and a scalar s,

$$v + s = [v_1 + s,\ v_2 + s,\ \ldots,\ v_n + s]$$

3. **Example:**

$$[2, 2] + 5 = [7, 7]$$
$$[2, 3] + 3 = [5, 6]$$

Subtraction works the same way, with s subtracted from every component:

$$[2, 3] - 1 = [1, 2]$$

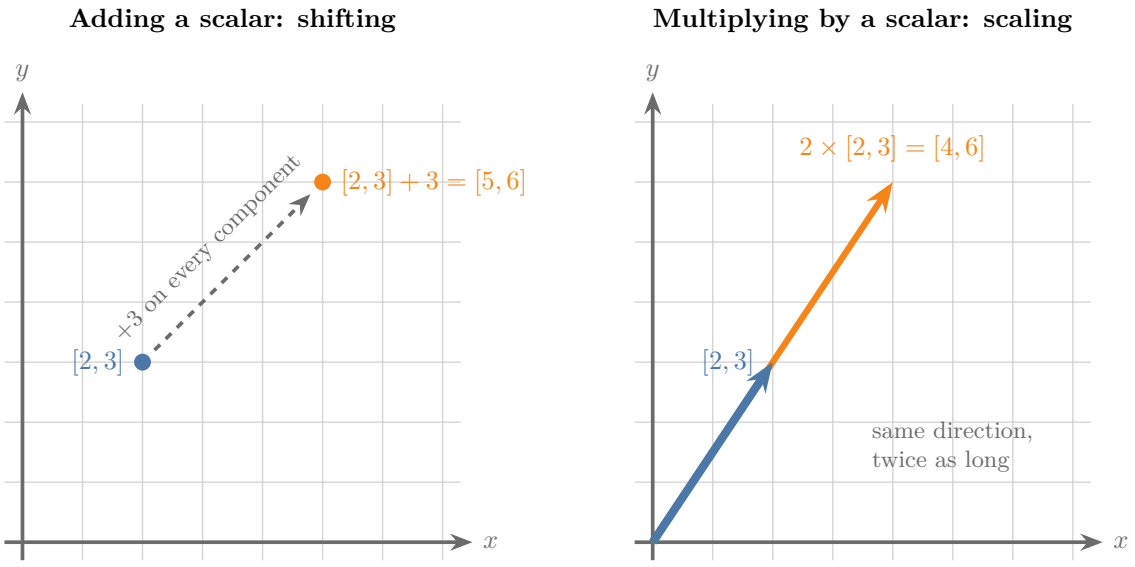


Figure 4: Left: adding a scalar moves the point. Right: multiplying by a scalar stretches the arrow

Changing every component moves the point to a new place in the coordinate system (Figure 4, left). So adding or subtracting a scalar is called **shifting** (G-1791).

> **Extra:** Strictly, adding a scalar to a vector is not an operation of linear algebra; a vector space has only two operations, adding two vectors and multiplying a vector by a scalar (MML Definition 2.9). NumPy allows it through **broadcasting** (G-333): in the words of the NumPy user guide (NumPy, "Broadcasting"), the scalar is "stretched" into an array

of the same shape, here the vector $[s, s, \ldots, s]$, and that is added. So $[2, 3] + 3$ is really $[2, 3] + [3, 3]$.

5. Mean centring: shifting in ML

Key point: Mean centring is a **scalar** (G-1743) subtraction on vectors: each feature minus its own mean.

Mean centring (G-1195), the first half of standardization (see what standardization does to the data), slides the cloud of points until its mean sits at the origin, without changing its shape. As a vector operation, mean centring is **shifting** (G-1791). Each **feature** (G-772; an input variable, one column of the data table) is a vector of values, and its mean is a scalar. Mean centring subtracts that scalar from every component, the way moving every house on a street by the same distance leaves the street's shape unchanged. For the feature $[3, 5, 7]$, with mean 5:

$$[3, 5, 7] - 5 = [-2, 0, 2]$$

Figure 5 does this to the 150 iris flowers, with their sepal length and sepal width as two features. Every flower moves by the same vector, minus the mean $(5.84, 3.06)$, so the cloud slides as one piece until its mean sits on the origin; its shape and spread do not change.

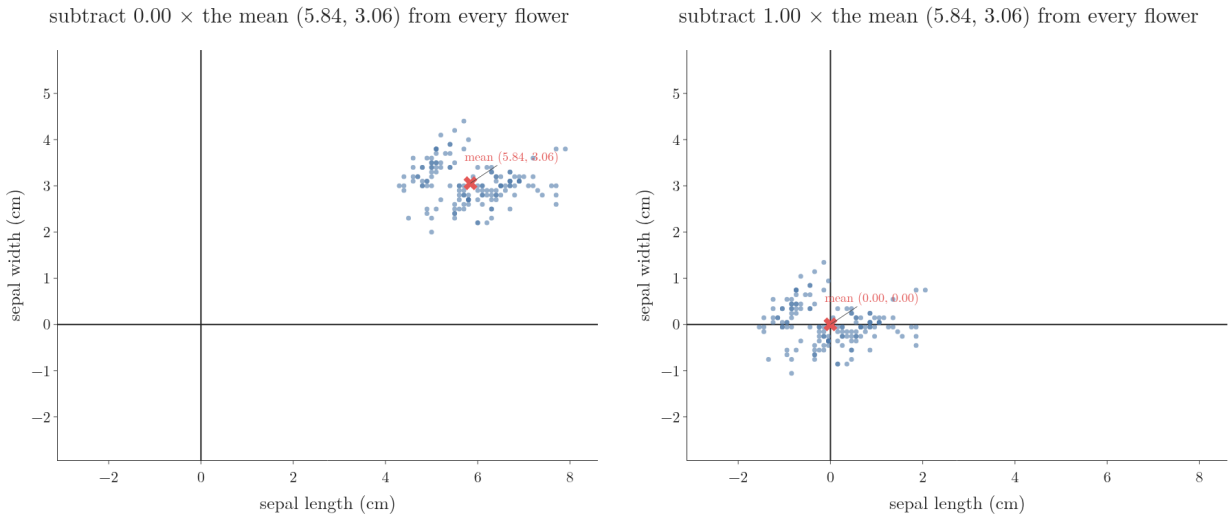


Figure 5: The 150 iris flowers (sepal length and width) shifted by a growing fraction of their mean, until the mean (red cross) sits at the origin. The cloud keeps its shape.

Python: Mean centring is one line: subtracting a row of means from a table subtracts each mean from its own column.

```
data = np.array([[3, 4], [5, 6], [7, 8]])
centred = data - data.mean(axis=0)    # [[-2,-2],[0,0],[2,2]]
```

`axis=0` takes the mean down each column. The Notebook for this Note

(`MA-049-magnitude-distance-and-scalar-operations.ipynb`) runs every example of this Note, plus a 100-vector mean-centring plot and the nearest-neighbour demo with a query you can change.

6. Multiplying or dividing by a scalar: scaling

Key point: Multiplying by a scalar multiplies every component, which stretches or shrinks the arrow without turning it; a negative scalar also flips it.

The second scalar operation works like the first, with multiplication in place of addition. Figure 4 (right) and Figure 6 show it on the vector $[2, 3]$, whose length is $\sqrt{13} \approx 3.61$. We take the scalars one at a time.

1. **A scalar above 1 stretches.**

$$2 \times [2, 3] = [4, 6]$$

   The new arrow lies on the same line through the origin and points the same way, but is twice as long: 7.21.

2. **A scalar between 0 and 1 shrinks.**

$$0.5 \times [2, 3] = [1, 1.5]$$

   Same direction, half the length. Dividing by s is the same as multiplying by $1/s$, so this is also $[2, 3]/2$.

3. **The scalar -1 flips.**

$$-1 \times [2, 3] = [-2, -3]$$

   The length stays 3.61, but the arrow now points the opposite way.

4. **A negative scalar flips and scales.**

$$-2 \times [2, 3] = [-4, -6]$$

   The opposite direction, and twice as long.

The number line under Figure 6 shows why a negative scalar flips. Multiplying the number 1 by 2 moves it out to 2, on the same side of zero. Multiplying it by -1 sends it to -1, the same distance from zero on the other side. A vector behaves the same way along its own line through the origin.

The operation is called **scaling** (G-1746), and the name **scalar** (G-1743) comes from it: a scalar is a number that scales a vector.

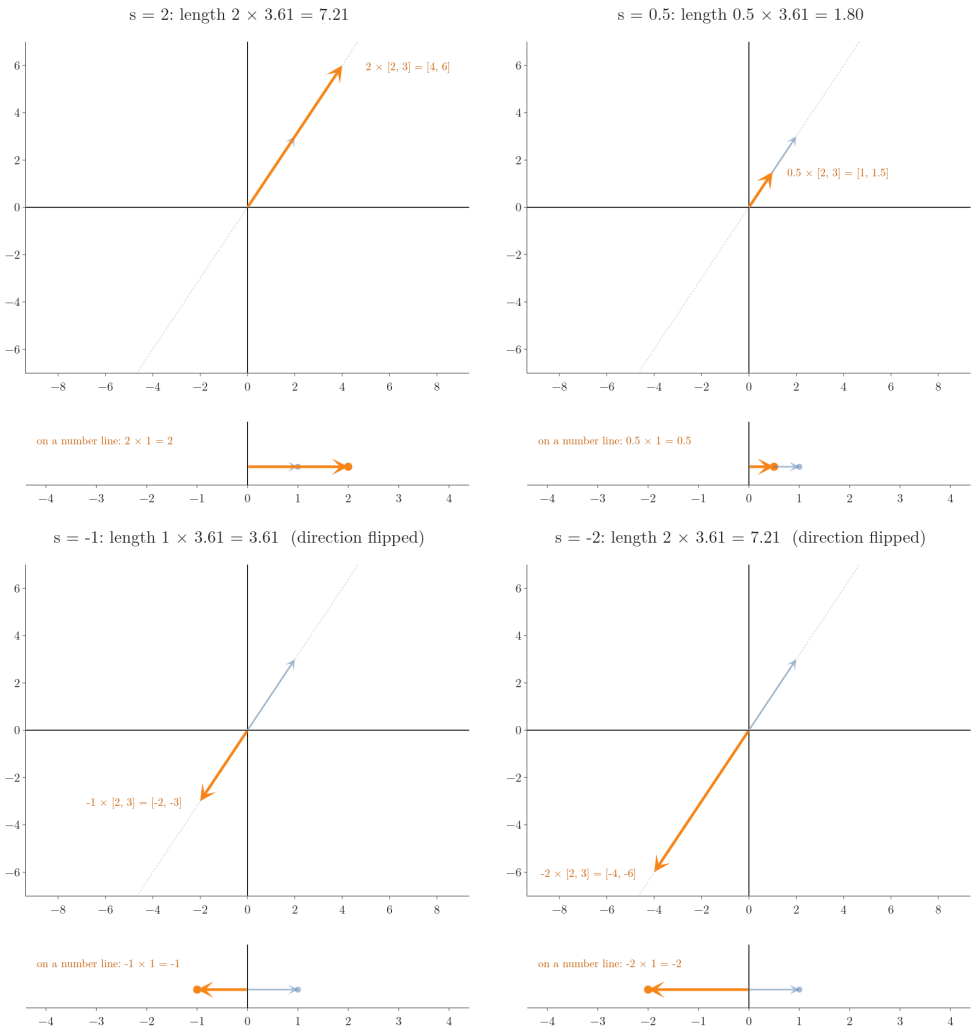


Figure 6: The vector [2, 3] (pale blue) multiplied by 1, 2, 0.5, −1 and −2 (orange). Every result lies on the same line through the origin; the length is multiplied by the size of the scalar, and a negative scalar flips the direction. Below, the same scalar acts on the number 1 on a number line. Number-line idea after Khan Academy, "Multiplying a vector by a scalar".

The formal version.

1. **In words:** multiply every component by the scalar.
2. **Formula:**

$$s\,v = [s\,v_1,\ s\,v_2,\ \ldots,\ s\,v_n]$$

3. **Effect on the length:** the magnitude is multiplied by $|s|$, the **absolute value** of s (its size with the sign dropped, so $|-2| = 2$ and $|0.5| = 0.5$). For $x = [2, 3]$ and $s = -2$, we get $sx = [-4, -6]$, with length:

$$\sqrt{16 + 36} = \sqrt{52} = 2\sqrt{13}$$

This is twice the length $\sqrt{13}$ of x. In general:

$$\|sx\| = \sqrt{s^2x_1^2 + \cdots + s^2x_n^2}$$

$$\|sx\| = |s|\,\|x\|$$

Dividing a vector by its own magnitude scales it to length 1, giving the **unit vector** (G-2048) used in projecting one point. For $[3, 4]$, whose length is 5:

$$[3, 4]/5 = [0.6, 0.8]$$

7. Summary

Idea	Formula	Example	Why it matters
Magnitude	$\|x\| = \sqrt{\sum x_i^2}$	$\|[3,4]\| = 5$	gives a length to vectors too big to draw; Ridge and Lasso penalise norms
Euclidean distance	$\|p - q\|$	$[1,1]$ to $[4,5]$: 5	KNN, K-means and recommenders decide by it
Shifting	$v + s$: add s to every component	$[2,3] + 3 = [5,6]$	moves the point to a new place
Mean centring	each feature minus its mean	$3,5,7 \to -2,0,2$	puts the data's mean at the origin without changing its shape
Scaling	$s\,v$: multiply every component by s	$2 \times [2,3] = [4,6]$; $-1 \times [2,3] = [-2,-3]$	changes the length, not the line; dividing by the length gives a unit vector

- Pythagoras' theorem gives both the length of a vector and the distance between two vectors, in any dimension, so a 50-component vector that cannot be drawn still has a length.
- Distance = magnitude of the difference: `np.linalg.norm(p - q)`, so one function measures both.
- KNN, K-means and recommender systems all decide by distance, because a new point takes the class, group or match of the vectors nearest to it.
- Shifting moves a vector, scaling stretches it; **mean centring** (G-1195) is shifting applied to whole features, so the data cloud slides to the origin as one piece.
- So one rule, Pythagoras, measures vectors in any dimension, and two **scalar** (G-1743) operations move and stretch them.

8. Sources

Built from

- CampusX, "Supercharge Your ML Journey: Mastering Vectors in Linear Algebra - Part 1", YouTube, https://www.youtube.com/watch?v=mQewAJb8oJ8
- Khan Academy, "Multiplying a vector by a scalar | Vectors and spaces | Linear Algebra | Khan Academy", YouTube, https://www.youtube.com/watch?v=ZN7YaSbY3-w
- Khan Academy, "Vector dot product and vector length | Vectors and spaces | Linear Algebra | Khan Academy", YouTube, https://www.youtube.com/watch?v=WNuIhXo39_k

Other references

- Deisenroth, M. P., Faisal, A. A. and Ong, C. S. (2020). *Mathematics for Machine Learning* (MML). Cambridge University Press. Definition 2.9 (Vector Space).
- NumPy developers. *NumPy User Guide*, "Broadcasting". numpy.org, basics.broadcasting.html.

9. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Mean centring (G-1195)	Subtracting the mean from every value, so the column's mean becomes 0.
L2 norm (norm, magnitude, length of a vector) (G-1028)	The usual magnitude of a vector, its distance from the origin: the square root of the sum of squared components, $\|w\| = \sqrt{w_1^2 + w_2^2 + \ldots}$.
L1 norm (G-1025)	The sum of the absolute values of the components.
Shifting (G-1791)	Adding or subtracting the same scalar to every component of a vector, such as $[2, 3] + 1 = [3, 4]$; it moves the point to a new place.
Broadcasting (G-333)	NumPy stretching a single number (a scalar) or a smaller array to match a bigger array before an operation.
Scaling (a vector) (G-1746)	Multiplying or dividing every component of a vector by the same scalar, such as $2 \times [1, 3] = [2, 6]$; it makes the arrow longer or shorter along the same line.
Scalar (G-1743)	A single number, such as 5 or -6: a 0D tensor; multiplying a vector by it scales the vector, which is where the name comes from.
Euclidean distance (G-715)	The straight-line distance between two points.
K-nearest neighbours (KNN) (G-998)	Predicting from the answers of the k closest stored points.
Feature (G-772)	One piece of information about each example that a model uses (e.g. a student's CGPA).
Unit vector (G-2048)	A vector of length 1, used to describe a direction.